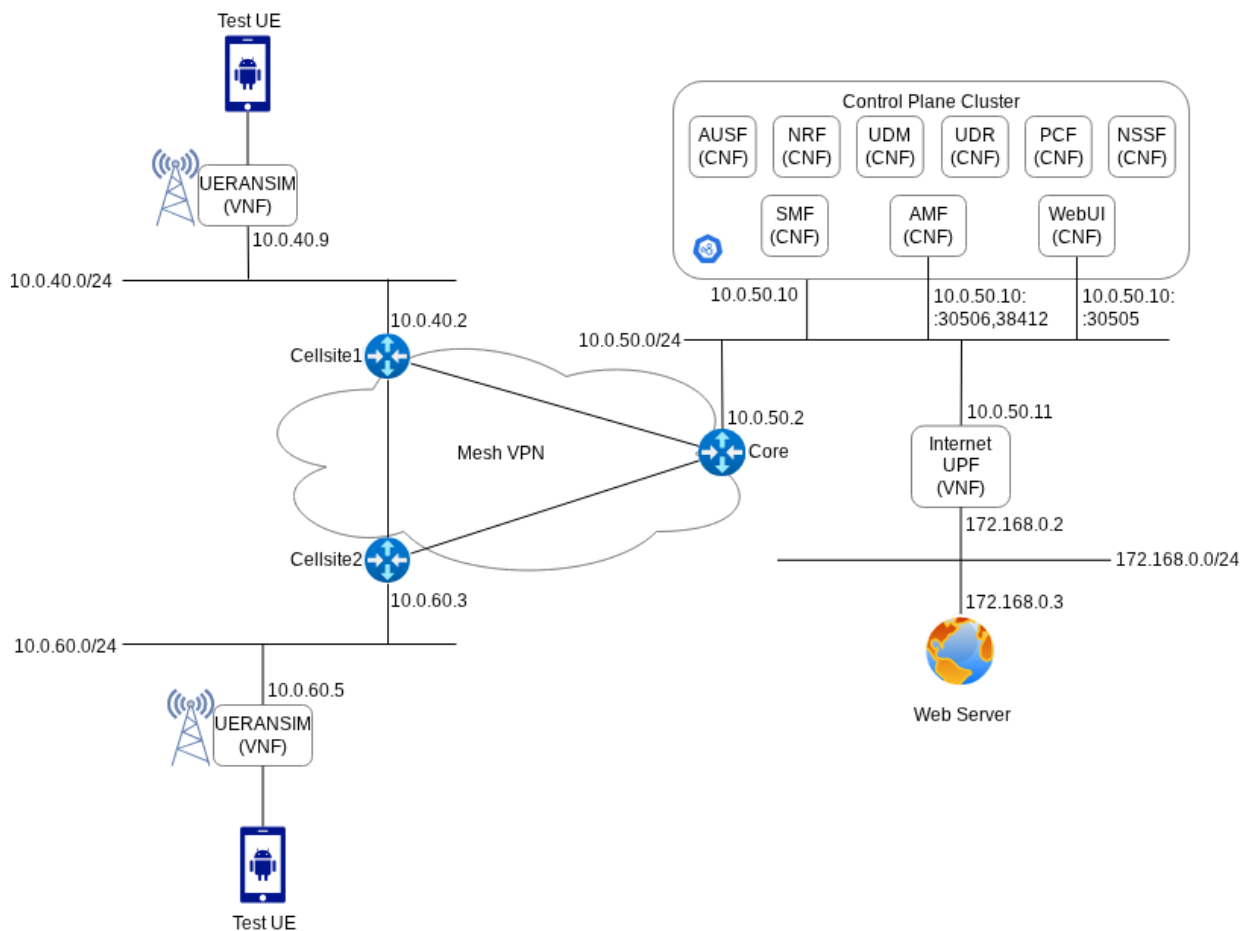


5G Network Design

A fully operational 5G network service will need one control plane, one user plane, one data network and one or more radio simulators to function correctly.



Network locations attached to UERanSim and UserPlaneFunction network services must be attached to a connectivity service so traffic can be carried between them.

When connecting more than two network locations you should use a Mesh connectivity service with multiple interfaces.

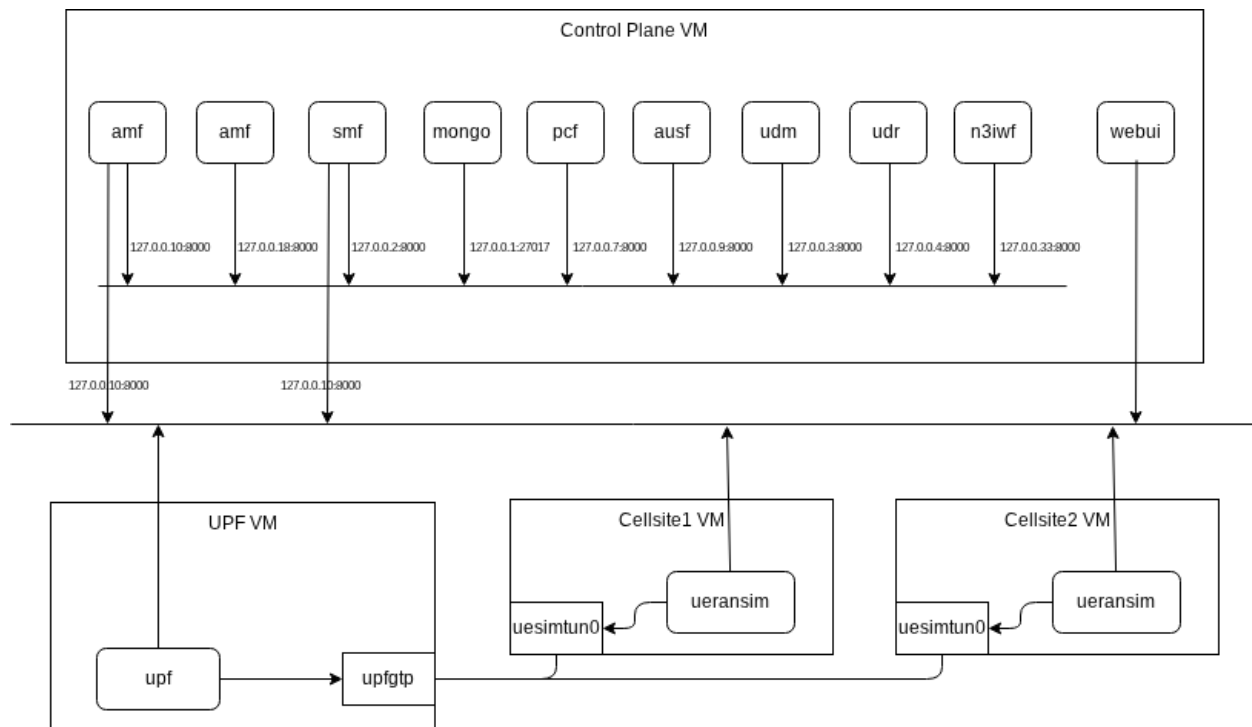
The following virtual infrastructure can be deployed to instantiate the 5G network service.

- **Core Network Site:** Based on Free5gc 5G Core network functions

- GCP networks are deployed to attach DNN mimicking the Internet and a Core network attaches control plane and upf network functions
- Free5gc Control Plane CNFs are deployed in a virtual machine attached to the core network.
- Free5gc UPF VNF is deployed attached to core and internet networks, routing between the two.
- **Radio Sites:**
 - GCP network per radio "site"
 - UERANSIM gNB Radio Network Simulator VNF is attached to the cellsite network. UE simulators can establish sessions and test traffic routed through the 5G network to the internet services above.
- **VPN Services:** Connecting all sites
 - Wireguard VNFs create a set of tunnels in a mesh or point to point configuration between GCP networks.

Once deployed test traffic can be run from the simulated UEs across the network to the Internet.

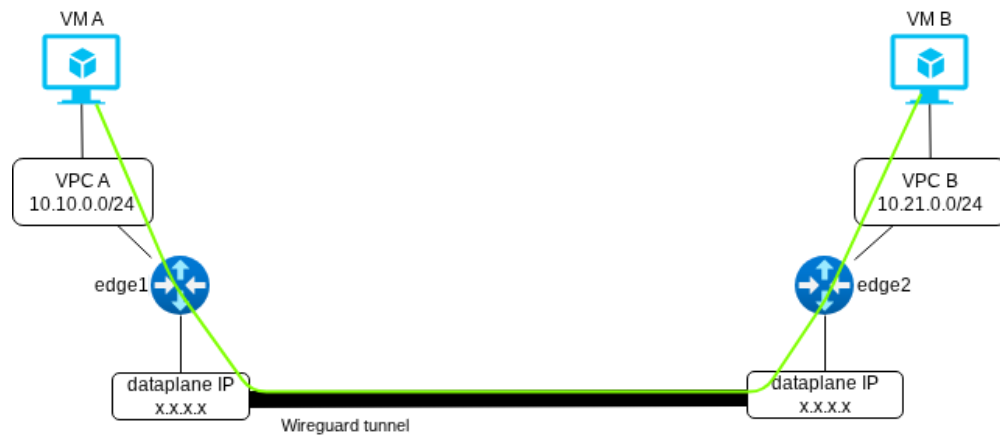
Free5gc Network Service



VPN Services

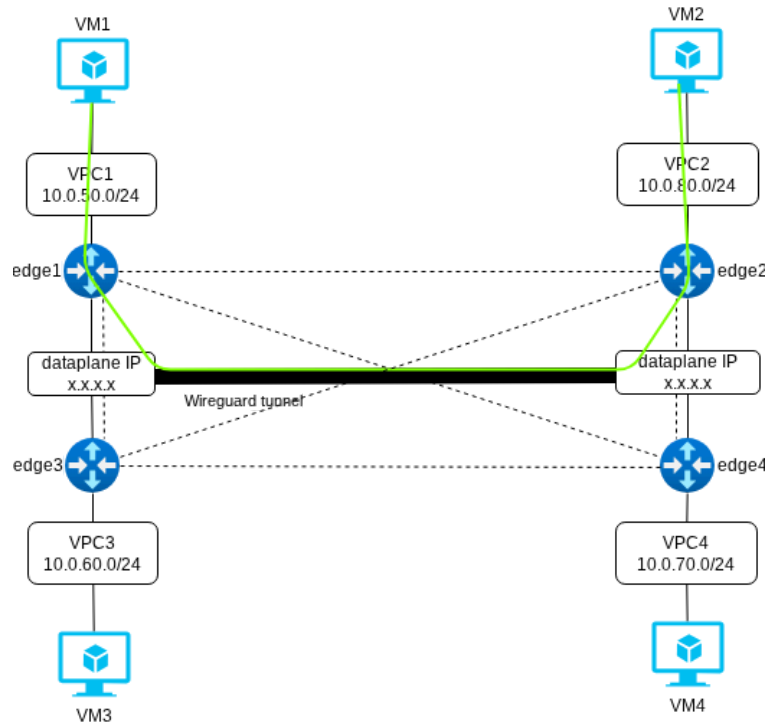
Point to point connectivity

This service connects two private VPCs over a wireguard tunnel. A pair of virtual network functions (VNFs) are connected to the two provided private VPCs. A wireguard tunnel is configured between the VNFs and static routes added to the private VPCs to route traffic over the VPN tunnel.



Mesh Connectivity

This service connects three or more VPCs over a set of wireguard tunnels. All VPCs can route traffic to/from the other VPCs in the Mesh.



IPAM

If you need to propose names for new network services or locations use the following guidelines:

- new network location names are at your discretion to propose
- When creating new network locations the following CIDR ranges are not to be used, i.e. these CIDRs are already used by the system
 - 10.0.0.0/24
 - 10.0.100.0/24
 - 10.60.0.0/24
- When creating new network locations check that the ip address with cidr for existing network locations
- new network service names are at your discretion
- new connectivity service names are at your discretion
- UserPlaneFunction and UERanSim network services must not be assigned the same network locations.
- The network location assigned to DataNetwork network service should be the same as the network location assigned to the UPF network location
- the dataplane network location is a reserved network location, you must not use it in your planned steps.